CS 6530 Applied Cryptography

Course Orientation and Introduction to Applied Cryptography

Foundations and Threat Modelling

A QUICK CRYPTOGRAPHIC WARM-UP

UIF TFDSFU JT PVU UIFSF.

What changed?

What is secret?

What is assumed?

Do not claim “secure” until these questions are answered.

BREAKING THE CIPHER: THE FIRST LESSON

THE SECRET IS OUT THERE.

Readable again
≠ proven secure

- **Which security property must be protected?**
- Against which adversary?
- Under which assumptions?
- In which deployed context?

COURSE ORIENTATION AND INTRODUCTION

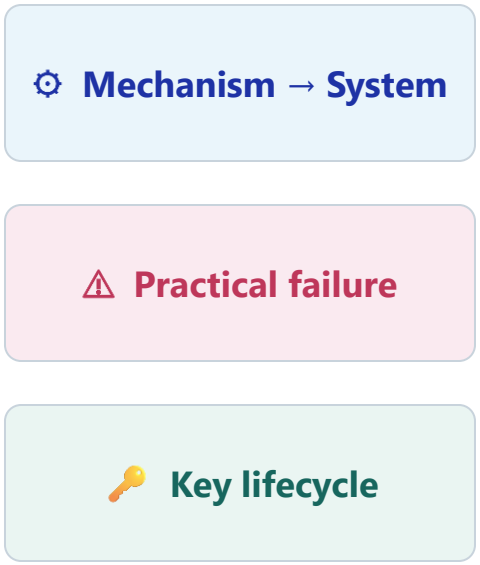
Applied cryptography connects mathematical mechanisms to protocols, implementations, key management, deployment choices and operational controls.

The recurring question is not only “Is the algorithm secure?”
It is also: “Does the complete system achieve the required security property?”

LEARNING OUTCOMES

By the end, you should be able to:

- explain cryptography's role in contemporary systems;
- distinguish primitive, construction, protocol, implementation and deployed system;
- identify where strong cryptography may fail in practice;
- explain why key management and deployment matter; and
- describe the course's applied learning approach.



CRYPTOGRAPHY IS ALREADY AROUND US

Choose one activity. Where might cryptography help—and what could still go wrong?

PHONE UNLOCK	**HTTPS**	**SOFTWARE UPDATE**	**BANK LOGIN**
MESSAGING	**PAYMENT**	**VPN**	**CLOUD FILE SHARING**

A WORKING DEFINITION OF APPLIED CRYPTOGRAPHY

Cryptography uses mathematical techniques to protect information.

Applied cryptography is the disciplined use of cryptographic mechanisms, protocols and key management to design, deploy, operate and evolve secure systems.

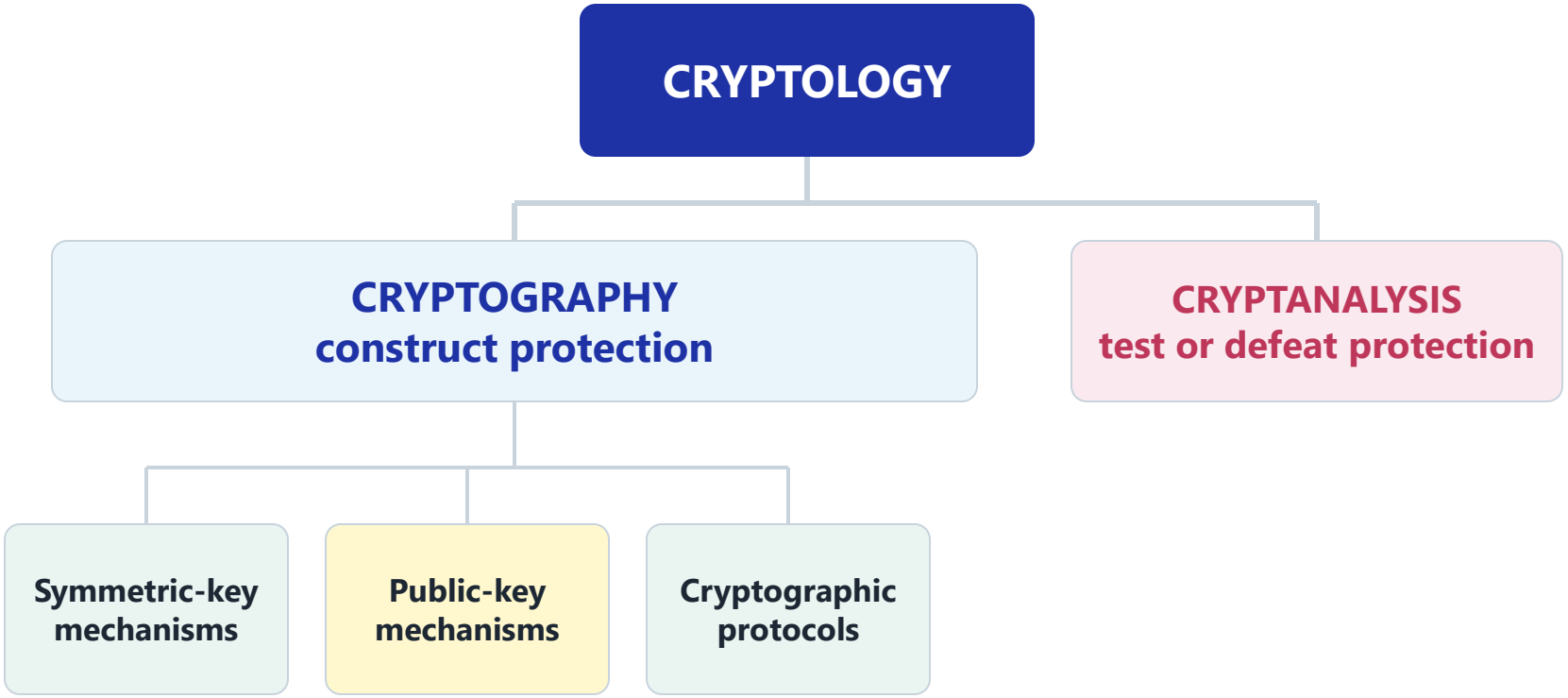
🛡 **Required property**

◉ **Adversary capability**

✓ **Explicit assumptions**

CRYPTOLOGY: CRYPTOGRAPHY AND CRYPTANALYSIS

A textbook classification of the field



CRYPTOGRAPHIC MECHANISMS SUPPORT DIFFERENT OBJECTIVES

Representative relationships—not exclusive one-to-one mappings.

ENCRYPTION	Confidentiality	**MAC**	Integrity + data-origin authentication
DIGITAL SIGNATURE	Integrity + origin authentication + evidence	**AUTHENTICATED ENCRYPTION**	Confidentiality + integrity/authentication
KEY ESTABLISHMENT	Establish shared keying material	**KEY MANAGEMENT**	Generate • store • distribute • rotate • revoke

ENCODING, ENCRYPTION AND HASHING

A changed appearance does not automatically mean protected information.

ENCODING	ENCRYPTION	HASHING
Format	**Confidentiality**	**Digest**
No secret required	Key required	Usually no secret

FIVE LEVELS FOR CRYPTOGRAPHIC SYSTEM ANALYSIS

A security claim changes meaning as we move from an algorithm to a deployed system.

5	**Deployed system**	identities • configuration • operations
4	**Implementation**	software or hardware
3	**Protocol**	message sequence and rules
2	**Construction**	primitives combined for a property
1	**Primitive**	AES • SHA-3

THE COMPOSITION PROBLEM

Security must survive every transition.



A strong first link does not repair a weak protocol, implementation or deployment.

MINI CLASSIFICATION ACTIVITY

At which level should each example be analysed?

AES	**ChaCha20-Poly1305**	**TLS**
Primitive	**Construction**	**Protocol**
Crypto library	**Enterprise VPN**	**Certificate authority**
Implementation	**Deployed system**	**Trust + operations**

ATTACKERS OFTEN BYPASS THE MATHEMATICS

Attackers often bypass strong cryptography rather than defeat it directly.



WHERE CAN THE SYSTEM FAIL?

A complete analysis checks every layer—not only the cipher.

Problem definition	wrong asset or property
Primitive / parameter	obsolete or unsuitable choice
Protocol composition	missing context, freshness or state
Implementation	leakage, randomness or verification error
Key management & operations	exposure, unsafe defaults or recovery

THE KEY HAS A LIFECYCLE

The algorithm protects data; the key controls that protection.



Every stage creates security requirements, operational responsibilities and failure modes.

A RECURRING ANALYSIS LENS

Use the same questions whenever you meet a new mechanism or protocol.

1 PURPOSE

Asset • property • guarantee

2 ADVERSARY

Observe • alter • replay • compromise

3 ASSUMPTIONS

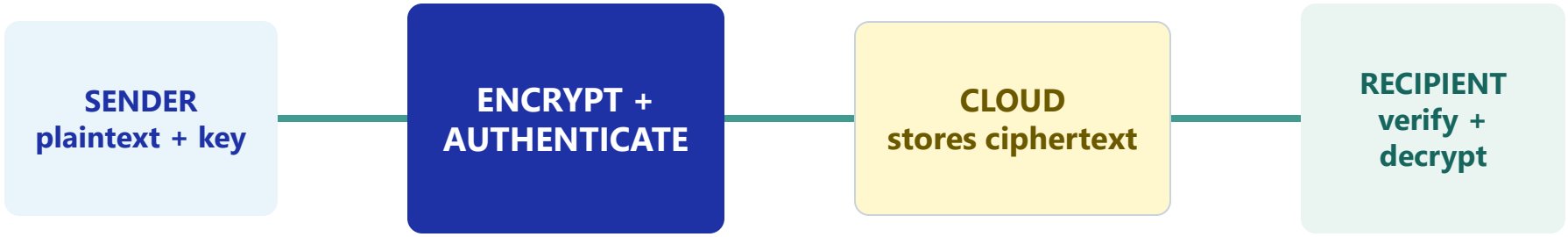
Keys • freshness • trust • parameters

4 VALIDATION

Implementation • configuration • operation • testing

INTRODUCTORY EXAMPLE—SECURE FILE SHARING

Goal: protect a confidential file during cloud-based sharing and detect modification.



System questions: key distribution • membership change • versions • backups • endpoints • metadata • replay

COURSE ORIENTATION FOR ENROLLED STUDENTS

The academic introduction is complete.

The following short segment explains how CS6530 is organized for enrolled students.

CLASSROOM ORIENTATION

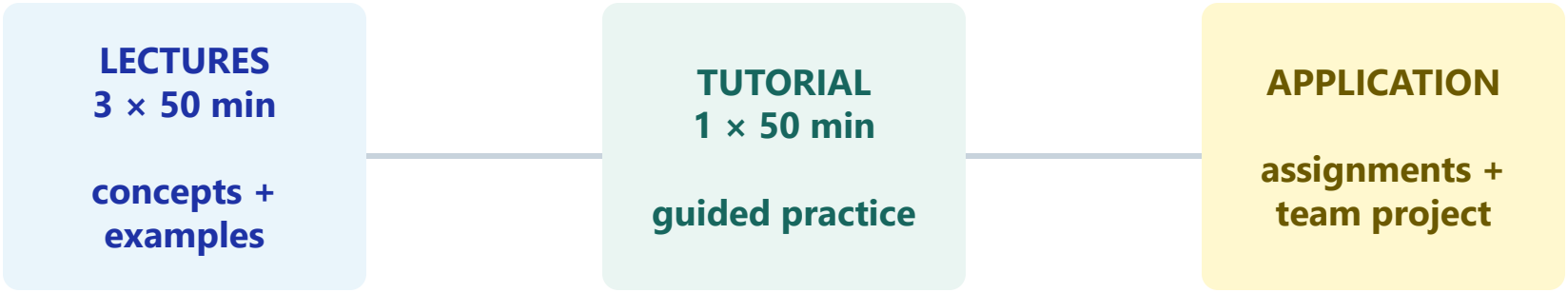
COURSE ROADMAP



The sequence is application-driven; references reinforce each topic.

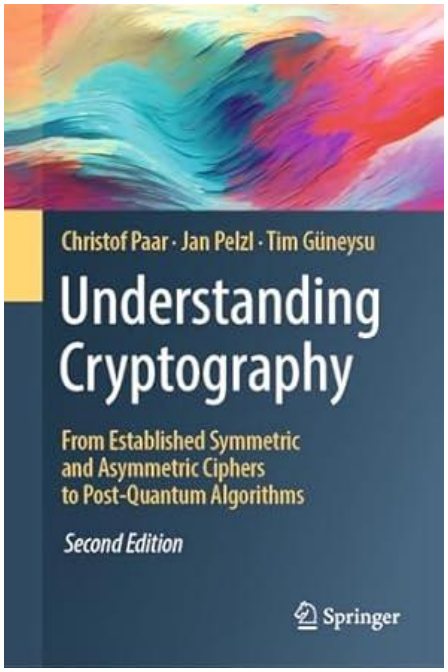
WEEKLY LEARNING PATTERN

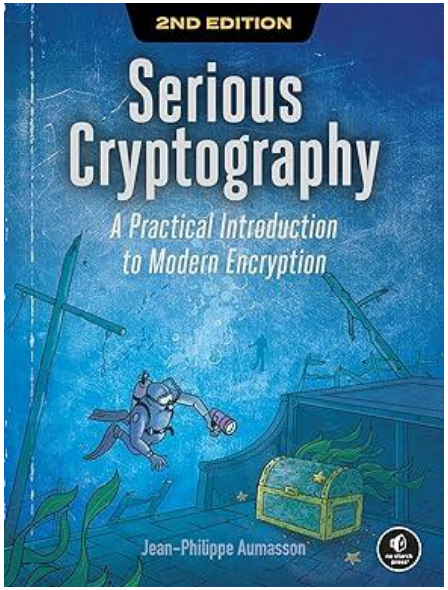
Each mode reinforces the same cycle of understanding, practice and application.



Notes + slides + tutorial practice form one connected learning resource.

PRINCIPAL REFERENCES AND TARGETED READING





FOR THIS TOPIC

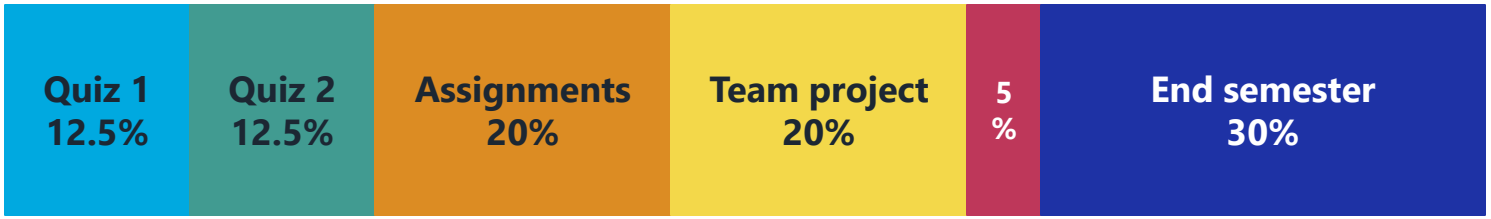
Understanding Cryptography
Chapter 1

Serious Cryptography
Chapters 1 and 3

Read after class to reinforce the conceptual foundation.

ASSESSMENT AND OPEN-NOTES EXPECTATION

Reasoning and justification matter. Some MCQs may have multiple correct answers.



Continuous Assessment (CA): 5% — best 5 out of 8 assessments count.

Open notes: specified printed and handwritten materials only.
Check the student handout for the complete policy.

STRONG CIPHER ≠ SECURE SYSTEM

EXIT QUESTION

How can strong encryption be present in an insecure system?